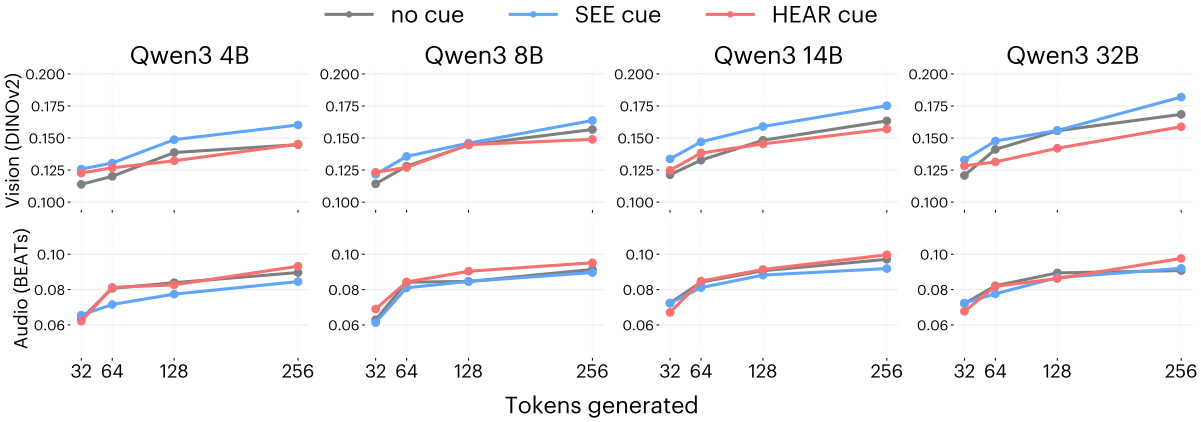


Figure 13: Extension of Figure 6 to "describe" instructed prompting.

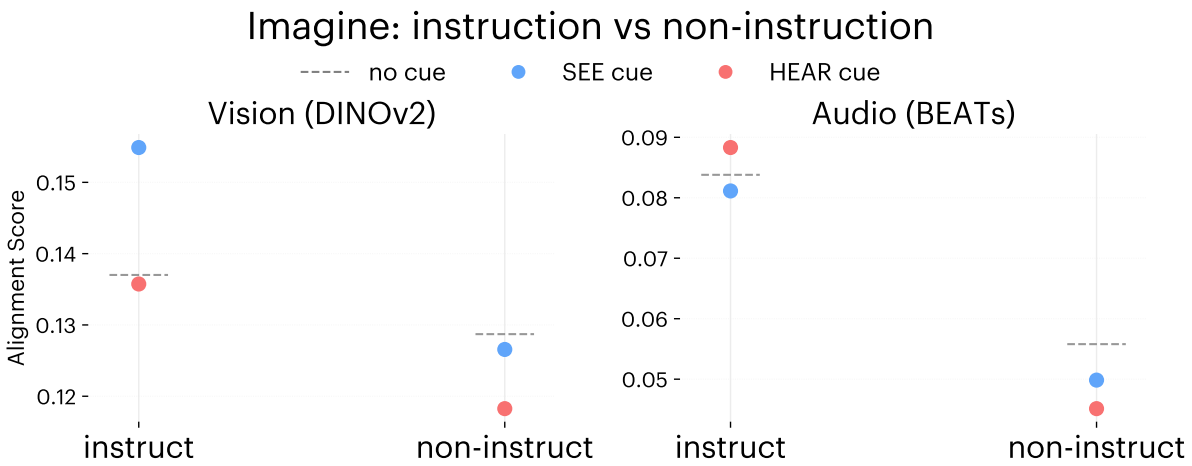


Figure 14: Alignment scores without instruction prompting.

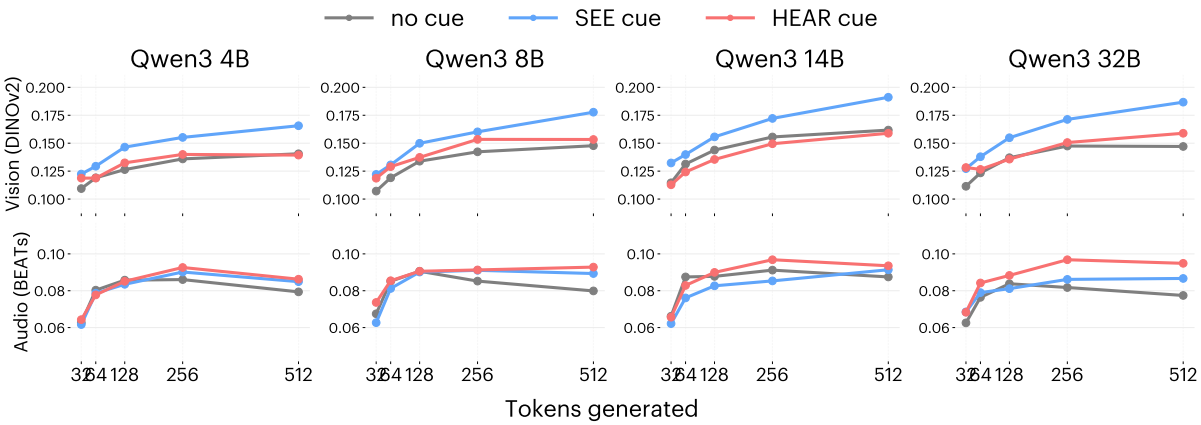
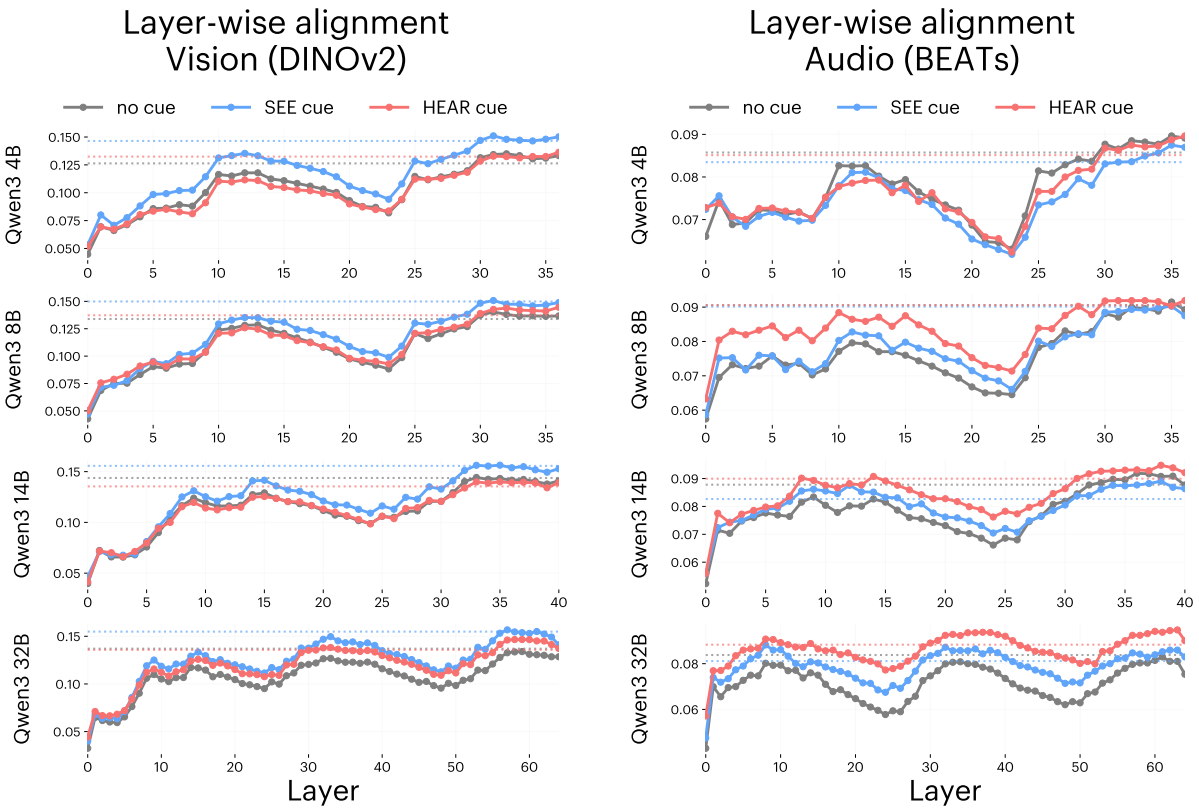


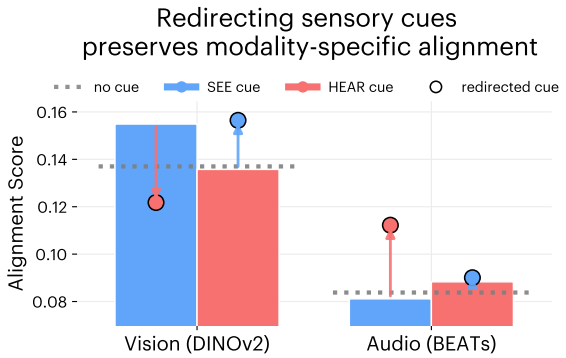
Figure 15: Extension of Figure 6 to 512-token generations.



WiT image–text dataset

AudioCaps audio–text dataset

Figure 16: Layer-wise alignment with 128-token prompts. The sensory prompting effect is consistent across layers. The dotted line indicates alignment when using the mean embedding across all layers.



(a) Redirection flips which modality the generations' representations align with.

Cue redirection

```
Please rewrite the following text to describe
what the scene would look/sound like visually/
audibly, rather than how it sounds/looks: {text}.
```

(b) Prompt template for redirection.

Figure 17: Redirecting sensory cues.

B Additional Qualitative Examples

Mutual-kNN alignment also lets us examine where sensory prompting increases or decreases overlap with the sensory encoder.

Figure 18 shows four WiT captions where mutual-kNN overlap with the vision encoder shifts under sensory prompting. For "Nasi goreng Pattaya...", (Section B.1) the no cue output highlights cultural context (*"Nasi goreng Pattaya is a local delicacy from Pattaya, Thailand, but it's also popular in neighboring countries"*), retrieving neighbors such as "Yam thale" (Thai dish), "Lankascincus gansi" (a skink species), and the "Korean–Chinese Cultural Center." With the SEE cue, the continuation instead describes visual details (*"the main components: fried rice, omelette, and the sauce… toppings like shrimp, chicken, or vegetables"*), and the neighbors are foods such as "Blinchiki filled with cheese and topped with blackberries" and "Spaghetti topped with pulled pork in a marinara sauce with a barbecue sauce base," yielding higher overlap with the vision encoder.

A similar shift occurs for the stinking beetle caption "Nomius pygmaeus..." (Section B.2). The no cue continuation paraphrases the encyclopedic description, producing neighbors that mix insects with unrelated entries such as "Lankascincus gansi" (a skink species), "The buttercup (Ranunculus spp) occurs in many variations…," and "Sawley Abbey, near to Sawley, Lancashire, Great Britain." With the SEE cue, the continuation instead emphasizes visible traits (*"a small, dark-colored beetle with a hard exoskeleton"*), and the neighbors become dominated by insect images such as "Anasimyia lineata" (hoverfly), "Pellenes seriatus" (spider), and "Winged caste of Huberia striata" (ant). This illustrates how visual prompting shifts the representation toward appearance-based similarity rather than encyclopedic associations.

In contrast, not all captions benefit from visual prompting. The bottom two rows show examples where visual prompting reduces alignment. The no cue continuation stays geographic, beginning with *"Okay, the user is asking about the 'Flag of ntice, Plze–South District.' First, I need to check if there's any existing information about a place called 'ntice' in the Plzeň–South District."* This framing retrieves sensible neighbors related to civic symbols such as "Nova tifta, Municipality of Sodraica, Slovenia," "Flag of Czech village Ln," and "Coat of arms of ievac." With the SEE cue, however, the continuation drifts into speculation about spelling and fictional names: *"Maybe they meant 'notice'? But that doesn't make sense … or perhaps 'Ntice' is a fictional or misspelled name."* As a result, the nearest neighbors shift away from flags toward mismatched entries such as "Planjava Northwest, seen from," "Coat of arms of ievac," and "Wine cellar in Szld." Here, visual prompting lowers overlap with the vision encoder by pulling the representation away from concrete geographic symbols and toward unrelated objects and places.

For the caption "MOLA map of Suess" (Section B.4), the no cue continuation stays on-topic, explaining that *"MOLA stands for Mars Orbiter Laser Altimeter … it created topographic maps of Mars by measuring the time it takes for a laser pulse to reflect off the surface and return to the spacecraft."* This technical framing retrieves neighbors tied to Mars imagery such as "This topographic map is created using Mars Orbiter Laser Altimeter (MOLA) technology … Cerulli crater," "Troughs and streaks in Arcadia quadrangle, as seen by hirise under HiWish program," and "King Lear Peak from Sulphur." By contrast, the SEE cue steers the continuation into speculative territory, with text like *"Suess is likely referring to the fictional land of Narnia … maybe the planet from The Hitchhiker's Guide to the Galaxy … or Dr. Seuss, the author."* The resulting neighbors ("Grotheer in 2018," "Planjava Northwest, seen from," "Daggett in 1984") lack clear visual connection to Martian maps. Here, visual prompting reduces overlap with the vision encoder by shifting the representation away from genuine topographic descriptions and toward unrelated associations.

Together, these examples highlight that sensory prompting increases alignment when it elicits modality-relevant descriptors, but can hurt when the cue introduces ambiguity or distracts from the grounded semantics of the caption.